student_mat

Importerer først alle bibliotekene jeg trenger, og filen vi skal jobbe med

```
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.3
```

```
Warning: package 'ggplot2' was built under R version 4.5.3
```

```
Warning: package 'tibble' was built under R version 4.5.3
```

```
Warning: package 'tidyr' was built under R version 4.5.3
```

```
Warning: package 'readr' was built under R version 4.5.3
```

```
Warning: package 'purrr' was built under R version 4.5.3
```

```
Warning: package 'dplyr' was built under R version 4.5.3
```

```
Warning: package 'stringr' was built under R version 4.5.3
```

```
Warning: package 'forcats' was built under R version 4.5.3
```

```
Warning: package 'lubridate' was built under R version 4.5.3
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
```

```
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tidymodels)
```

```
Warning: package 'tidymodels' was built under R version 4.5.3
```

```
── Attaching packages ─────────────────────────────────────── tidymodels 1.5.0
──
✔ broom        1.0.13     ✔ rsample      1.3.2
✔ dials        1.4.4      ✔ tailor       0.1.0
✔ infer        1.1.0      ✔ tune         2.1.0
✔ modeldata    1.6.0      ✔ workflows    1.3.0
✔ parsnip      1.6.0      ✔ workflowsets 1.1.1
✔ recipes      1.4.0      ✔ yardstick    1.4.0
```

```
Warning: package 'broom' was built under R version 4.5.3
```

```
Warning: package 'dials' was built under R version 4.5.3
```

```
Warning: package 'scales' was built under R version 4.5.3
```

```
Warning: package 'infer' was built under R version 4.5.3
```

```
Warning: package 'modeldata' was built under R version 4.5.3
```

```
Warning: package 'parsnip' was built under R version 4.5.3
```

```
Warning: package 'recipes' was built under R version 4.5.3
```

```
Warning: package 'rsample' was built under R version 4.5.3
```

```
Warning: package 'tailor' was built under R version 4.5.3
```

```
Warning: package 'tune' was built under R version 4.5.3
```

```
Warning: package 'workflows' was built under R version 4.5.3
```

```
Warning: package 'workflowsets' was built under R version 4.5.3
```

```
Warning: package 'yardstick' was built under R version 4.5.3
```

```
── Conflicts ─────────────────────────────────────── tidymodels_conflicts()
──
✖ scales::discard() masks purrr::discard()
✖ dplyr::filter()   masks stats::filter()
✖ recipes::fixed()  masks stringr::fixed()
✖ dplyr::lag()      masks stats::lag()
✖ yardstick::spec() masks readr::spec()
✖ recipes::step()   masks stats::step()
```

```
library(workflows)
library(recipes)


#bruker delim, da jeg spesifiserer at cellene er delt med ;
data <- read_delim("student-mat.csv", delim = ";")|>
  rename(grades = G3)|>
  dplyr::select(age,Mjob,Fjob,traveltime,studytime,failures,absences)|>

  #gjør alle strenger til faktorer. Tall beholdes som doubles
  dplyr::mutate(across(all_nominal(), as.factor))
```

```
Rows: 395 Columns: 33
── Column specification
────────────────────────────────────────────────────────
Delimiter: ";"
chr (17): school, sex, address, famsize, Pstatus, Mjob, Fjob, reason,
guardi...
dbl (16): age, Medu, Fedu, traveltime, studytime, failures, famrel,
freetime...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```